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Investigating the Fusion Gap in Bidirectional Federated Distillation Systems



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Statutory Declaration

I hereby declare under oath that the submitted Bachelor's thesis titled "Investigating the Fusion Gap in Bidirectional Federated Distillation Systems" has been carried out by me in the institute of Pervasive Computing. The information derived from the literature has been duly acknowledged in the text and a list of references provided. This thesis has not previously been submitted, in whole or in part, to any other degree or diploma at this or any other institution.

Statement on the Use of Artificial Intelligence.

Large Language Models (Google Gemini 3.1 Pro, Google Gemini 3.5/3.6 Flash) have been utilized in preparation for this thesis for assistance with visualizations, structural planning of the thesis, and proofreading. The final content, research methodology, and all experimental analysis remain entirely my own work.

Linz, July 30, 2026	Vsevolod Kharchenko
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Abstract

Federated learning allows for decentralized model training on user data while maintaining data privacy. In real deployment, however, such issues as model heterogeneity and local data heterogeneity (Non-IID data) arise. Such frameworks as FedDF use knowledge distillation with a proxy dataset on the server to mitigate those issues, but the issues remain at high levels of data heterogeneity, and both server and client models end up having worse accuracies due to severe catastrophic forgetting. The clients are also observed to retain stability while the server model performance suffers, however, resulting in a phenomenon defined in this paper as the fusion gap.

This thesis empirically investigates the behavior of the fusion gap under different data heterogeneity scenarios and federated learning scenarios: regular FedDF and BiFedDF - FedDF modified with a bidirectional distillation component. The results reveal that BiFedDF trades server accuracy for much better client accuracy due to the bidirectional distillation component acting as a regularizer for client models, mitigating the catastrophic forgetting - loss of classification accuracy for classes underrepresented in the local dataset - and drastically reduces the difference in accuracy between the server and clients.

This improvement, however, comes with a reduction in client model diversity, thus lowering the effectiveness of an ensemble of client models, used to train the server model. This results in a worse server performance, and strengthens the fusion gap, especially as the level of data heterogeneity increases. Overall, this thesis shows that bidirectional distillation in federated learning comes at a trade-off between global server accuracy and local client stability.

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List of Acronyms

BiFedDF	Bidirectional Federated Distillation Fusion
CE	Cross-Entropy
CNN	Convolutional Neural Network
CPU	Central Processing Unit
DL	Deep Learning
FedAvg	Federated Averaging
FedDF	Federated Distillation Fusion
FL	Federated Learning
GAP	Global Average Pooling
GDPR	General Data Protection Regulation
GPU	Graphics Processing Unit
IID	Independent and Identically Distributed
IoT	Internet of Things
JSD	Jensen-Shannon Divergence
KD	Knowledge Distillation
KL	Kullback-Leibler
ML	Machine Learning
RAM	Random Access Memory
RGB	Red Green Blue
SGD	Stochastic Gradient Descent

Chapter 1

Introduction

1.1 Motivation

The personal motivation behind this thesis and the reason behind why I got into Knowledge Distillation and Federated Learning comes from my tendency to focus on the small details of the projects I was in during their later stages, or, in other words, the personal need to optimize everything within my reach (e.g., rewriting a model's architecture over and over to achieve a 0.5% accuracy increase and better readability).

The novelty of Knowledge Distillation as a Model Compression approach, as described in the Model Compression Survey by Li et al. [17], sparked my curiosity, and in combination with the convoluted scenarios of federated learning and the constraints of Edge devices, I chose to research Knowledge Distillation in Federated Learning.

1.2 Problem Statement

Due to the widespread adoption of IoT and smart edge devices, large amounts of decentralized data have emerged, which is useful for Deep Learning (DL) and other machine learning (ML) applications. Centralized storage and collection of user data, which is common when working with ML/DL, is problematic due to bandwidth limitations, many regulations (e.g., GDPR), and general privacy concerns.

Federated Learning (FL) aims to address this: instead of collecting the data and bringing it to one centralized server for model training, it sends the model to the data, eliminating the issue of centralization. In many scenarios, however, the users have their own models, much different from the server model in architecture. Additionally, the data in the local datasets is usually non-IID, leading most models to struggle with it, leading to client drift: local models optimizing for their own skewed data and diverging from each other heavily.

Knowledge Distillation [6] was used in the first successful heterogeneous FL methods and nicely mitigated client drift. FedMD (Li et al.) [16] was the first answer to heterogeneous federated learning. The client logits are trained on the local datasets and sent to the server, averaged, and then the clients are trained to approach the averaged logits via knowledge distillation. The heterogeneity issue is resolved at the cost of no longer having a server model, as the only thing done on the server is logit averaging from a decentralized ensemble of client models. FedDF (Lin et al.) [18] takes a different approach: now the clients are simply trained on their local sets, and a server model is trained to follow the averaged student logits. Unlike FedMD, it does have a server model, but its performance struggles with high data heterogeneity, despite one of the issues it set out to solve being heterogeneous data in federated learning, and the client models do not benefit from the aggregated global knowledge.

1.3 Thesis Statement

In my practical work, I have addressed the limitations of FedDF by modifying it so that the knowledge is also distilled back into the client models, resulting in the Bidirectional Federated Distillation Fusion (BiFedDF) framework. While it resulted in good client performance, it came at the cost of server performance. I have defined the server accuracy loss despite the average client accuracy's stability as the fusion gap.

This thesis aims to investigate the effect of the Bidirectional Distillation component of BiFedDF on the Fusion Gap under different degrees of data heterogeneity.

1.4 Research Questions

To investigate the causes of the fusion gap within BiFedDF, the following research questions were formulated:

1. How does the degree of data heterogeneity across clients affect the magnitude and behavior of the fusion gap in a bidirectional knowledge distillation FedDF system?
   - How does the global server model's performance change when transitioning from homogeneous IID data to increasingly heterogeneous Non-IID data?
   - How does the stability of the average client model accuracy change with increasing data heterogeneity?
2. Does the bidirectional distillation component — where clients also learn from the server — reduce or amplify the fusion gap compared to a standard unidirectional FedDF baseline, and does this relationship change under Non-IID conditions?
   - How does the severity of the fusion gap in BiFedDF quantitatively compare to FedDF?
   - How much does either framework outperform the other in terms of fusion gap mitigation with increasing data heterogeneity?

1.5 Objective of the Thesis

The main objectives and contributions of this thesis are the formal definition of the fusion gap in federated distillation and the insights into how bidirectional distillation affects federated learning performance compared to a FedDF baseline when heterogeneous data is involved.

1.6 Scope of the Thesis

The scope of the thesis is focused on the empirical evaluation of the fusion gap in FedDF and BiFedDF federated learning frameworks. The experiments consist of exclusively simulated federated learning scenarios using a dataset common in literature - CIFAR-10 (Krizhevsky et al.) [14] - distributed with different levels of data heterogeneity (IID, mild non-IID, strong non-IID), and no deployment on physical edge hardware occurs. Classification accuracy, client drift, and quantitative difference between server and client performance are evaluated. Other model compression techniques are not explored in this thesis outside the context of the distillation pipeline.

1.7 Thesis Structure

Background and related work (section 2): Describes the prior work that led up to this thesis, as well as the researched relevant literature. Introduces the base concepts and methods this thesis works with.

Methodology (section 3): Describes all methods, tools, frameworks and metrics used to conduct the experiments. Explains the federated learning frameworks used, the model architecture, and the data heterogeneity scenarios.

Experiment results (section 4): Shows the quantitative results of the experiments, includes a discussion on what they mean and how they can answer the research questions.

Conclusion and future work (section 5): Summarizes the conclusions and findings of this thesis. Suggests ideas for how to expand on this thesis and its findings.

Chapter 2

Background and Related Work

For my practical work, I was comparing knowledge distillation with other model compression techniques for federated learning. The idea for the project came from a 2023 Model Compression Survey by Li et al. [17].

2.1 Data Heterogeneity

An important thing to cover before continuing is **heterogeneous (or Non-IID) data**, as it's one of the things federated distillation was created to work with. In real scenarios, client data is rarely IID, and it's most likely unbalanced in terms of class representation. This leads to what is called client drift: models trying to optimize their parameters for skewed data. When these models are aggregated to the server, they conflict with each other, which will lower the server performance in most cases (Kairouz et al.) [10].

Researchers usually simulate data heterogeneity to ensure a controlled environment for their experiments. The most common method to do so is by using the Dirichlet distribution (Hsu et al.) [7], where the severity of data heterogeneity can be adjusted with a single parameter.

2.2 Model compression and Knowledge Distillation

Knowledge distillation (Hinton et al.) [6] is a way to transfer knowledge from a bigger model into a smaller one(figure 2.1). The way it works is by taking a large, pretrained teacher model and training a smaller model on the teacher's logits, typically with Kullback-Leibler divergence loss combined with cross-entropy loss:

$$\mathcal{L}_{KL+CE} = \beta \mathcal{L}_{KL} + (1 - \beta)\mathcal{L}_{CE}$$

$\mathcal{L}_{KD}$ = KLDiv loss, $\mathcal{L}_{CE}$ = Cross-Entropy loss, β - weights of the two loss functions

Kullback-Leibler Divergence measures the difference between some probability distribution (Q) and a reference distribution (P):

$$D_{KL}(P \parallel Q) = \sum_{i=1}^{C} P(i) \log \left(\frac{P(i)}{Q(i)} \right)$$

Combined with a temperature parameter (T) and a softmax function, we get the KLDiv Loss function:

$$\mathcal{L}_{KL} = T^2 \cdot D_{KL}\Big(\sigma(z^{teacher}, T) \parallel \sigma(z^{student}, T)\Big)$$

σ = softmax function

A softmax function is used to extract dark knowledge from the teacher - relative probabilities of incorrect classes. The function is modified by the Temperature parameter T:

$$\sigma(z, T)_i = \frac{\exp(z_i/T)}{\sum_{j=1}^{C} \exp(z_j/T)}$$

Temperature determines how much dark knowledge gets transferred: the higher it is, the more dark knowledge is shared, and the lower it is - the more the distribution will resemble a one-hot encoding.

A way in which knowledge distillation can be used to compress a model is by creating a smaller version of the model and distilling the knowledge from a smaller model into it. For good accuracy, however, it is best to define a model architecture that is not less than 2-4x the original model's size, according to Mirzadeh et al. [20]. In my practical work, however, I was using a student model over 40 times smaller than the teacher to see how far I could take model compression.

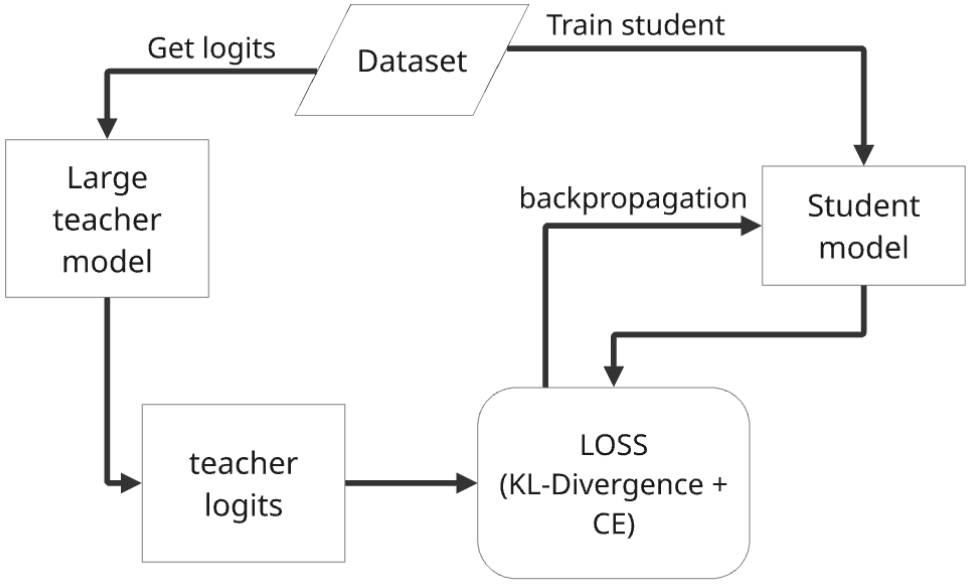


Figure 2.1: Knowledge Distillation

Initial scope of the practical work also covered other model compression techniques from the survey by Li et al. [17] such as:

- **Unstructured pruning** [3]: removal of n% of smallest weights in a model.
- **Quantization** [9]: reduction of numerical precision of the model (e.g. INT8 post-training quantization: a reduction of precision from 32-bit to 8-bit without pretraining).

However, this thesis will only focus on Knowledge Distillation in the context of model compression.

2.3 Federated Learning

Federated Learning is training models across multiple decentralized devices or nodes without needing to transferring data to server. The first federated learning framework defined by McMahan et al. [19] was FedAvg. The way it worked was by training clients locally, aggregating and averaging client weights at the server and then redistributing the averaged model to clients. This approach, however, did not support heterogeneous models and struggled with heterogeneous data, and therefore made knowledge distillation impossible, which is why I needed a different approach.

The most prominent methods that solved the issues of model and data heterogeneity were FedMD (Li et al.) [16] and FedDF (Lin et al.) [18], both of which used knowledge distillation to allow clients to have different model architectures, and both of which are state-of-the-art federated distillation approaches.

FedMD works by training the client models on local data, averaging their logits on the server, and training the client models to follow the averaged logits. While this allowed for heterogeneous models and data in Federated Learning, this framework did not have a server model. For this reason alone it had to be omitted in favor of FedDF. FedDF consists of training, where the client models are trained on their local data, and fusion, where the server model is trained to follow averaged client logits via knowledge distillation. It was created to deal with the issues of model and data heterogeneity in federated learning by using knowledge distillation to fuse and transfer knowledge from client models into a single server model. While it succeeds at that, the clients do not benefit from the model fusion. The FedDF framework is great for achieving great server performance, but not client performance.

2.3.1 Client Drift

A big flaw of FedAvg is the weight divergence under heterogeneous data, formalized by Zhao et al. [27]. They demonstrated that Non-IID data forces local models into divergent optima.

Such frameworks as FedProx (Li et al.) [23], in which a proximal term to the local objective function was introduced to mathematically restrict local weights from drifting too far from server weights, and SCAFFOLD (Karimireddy et al.) [11], which uses control variates to correct drift in local updates, were proposed to mitigate this client drift. While they were successful at stabilizing training, they did not support model heterogeneity. This is one of the limitations that paved the way for Knowledge Distillation in Federated Learning.

2.4 Bidirectional Distillation

Zhang et al. [26] introduced mutual distillation, also known as bidirectional distillation: taking knowledge distillation from teacher to student and applying it from student to teacher as well. I chose to apply it to FedDF to make the clients benefit from the server fusion. This way, I ended up with **BiFedDF** - Bidirectional Federated Distillation Fusion. During my experiments with BiFedDF, however, a weird phenomenon occurred: despite the stable client accuracies, the server model's performance kept dropping.

2.5 Bidirectional Federated Learning Methods

Several researches proposed using bidirectional distillation in federated learning to improve local client performance. I've researched other bidirectional federated learning methods in an attempt to find a solution to the issue I've experienced with BiFedDF:

- **FedBKD** (Qi et al.) [22]: federated bidirectional distillation with model heterogeneity support. Generates data for server and clients to train on. Data generation aside, very similar to what I did, but with a combined CE and KLDiv loss for client-to-server distillation. The scope of the paper is mostly on the data generation, which is why it does not describe anything similar to a global server degradation phenomenon.
- **FedBiKD** (Shang et al.) [24]: federated bidirectional distillation. Works with heterogeneous data, but not heterogeneous models. The paper mentions the local models achieving near perfect local test accuracy, but does not explore the effect this has on the ensemble gain.
- **FedDCM** (Huang et al.) [8]: decentralized federated learning. Supports both heterogeneous models and data, and uses a mix of KLDiv and CE loss for bidirectional distillation. As it's a decentralized method, it doesn't have a server, and therefore cannot have any issues similar to mine.

Chapter 3

Methodology and Experimental Setup

3.1 Research Approach and Objectives

The research methodology relies on a quantitative analysis, and the frameworks are not deployed on physical edge devices, but are simulated instead. This ensures control over experimental variables, such as the Dirichlet distribution α parameter. The analysis measures the classification accuracy of the server model and the average classification accuracy of the client models in 70 federated learning rounds. This way, we can produce plots and calculations to help us understand the effect each framework has on the fusion gap, which will help answer the research questions.

3.2 Federated Learning Approaches

In order to provide better understanding of the federated learning approaches used in the experiments, Federated Averaging or FedAvg (McMahan et al.) [19] must be explained first.

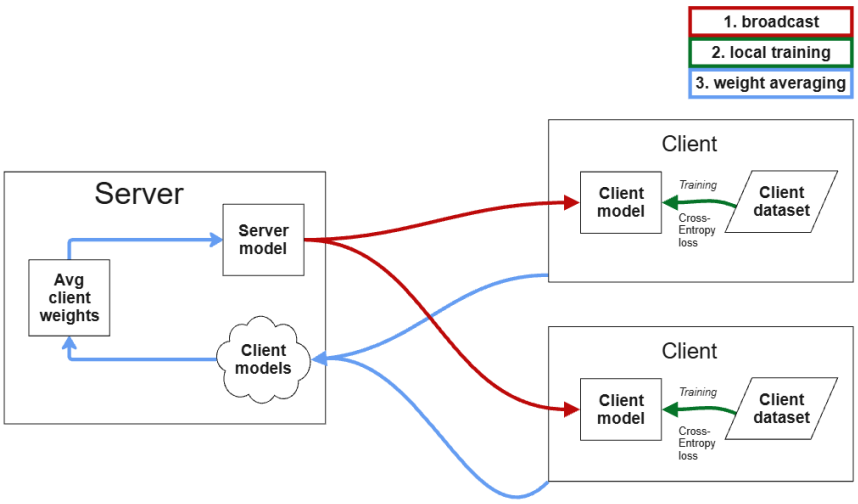


Figure 3.1: Federated Averaging (FedAvg): The server broadcasts the global model (red), clients perform local training on their private datasets (green), and the updated model weights are sent back to the server to be averaged and assigned to the server model (blue)

FedAvg is the first federated learning approach and thus serves as a foundation for all other federated learning approaches. It consists of 3 stages (figure 3.1):

1. Broadcast: the server model weights are sent to the clients to be used in the client models
2. Local training: training is performed on client models using local client datasets
3. Averaging: the client model weights are sent to the server, averaged, and assigned to the server model

This was the original intended approach for practical work. However, this approach did not support heterogeneous models, which made knowledge distillation unusable for model compression. Thus, the choice to use FedDF was used instead

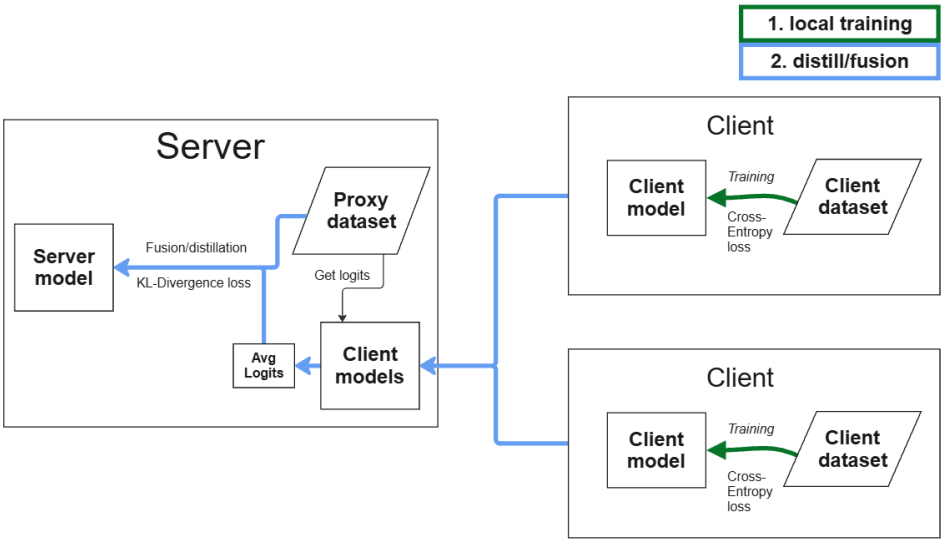


Figure 3.2: Federated Distillation Fusion (FedDF)

FedDF (figure 3.2) consists of two stages - training and fusion:

1. **Training**: the client models are trained on their local datasets
2. **Fusion**: the clients are sent to the server, and the server model is trained on their averaged logits.

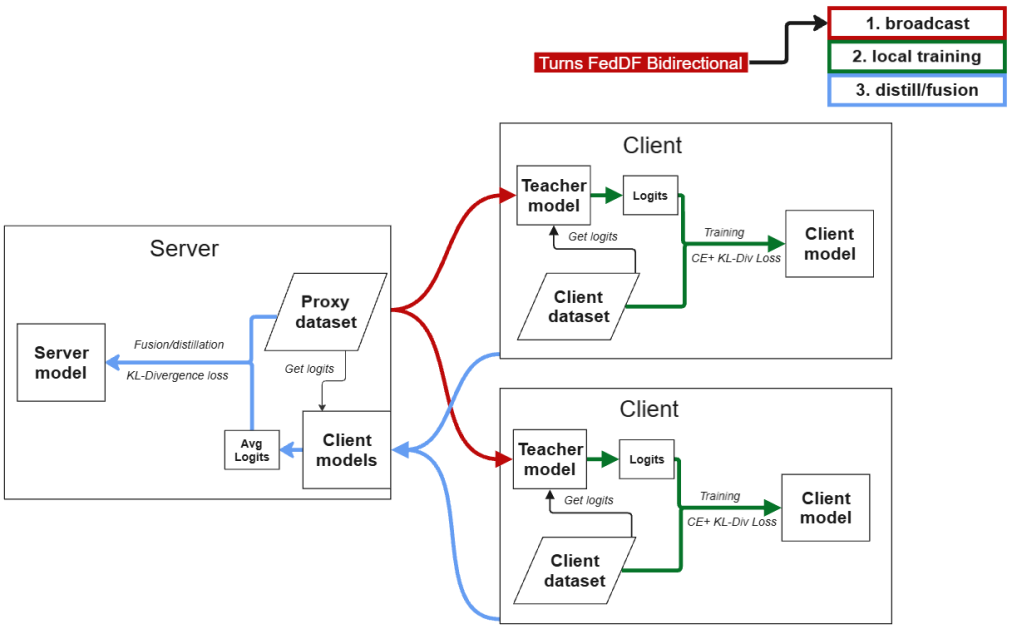


Figure 3.3: Bidirectional Federated Distillation Fusion (BiFedDF)

BiFedDF (figure 3.3) turns FedDF bidirectional. This is done by adding another step - **Broadcasting** the server model to the clients - and by training the clients to follow the server model along with regular training using a combination of cross-entropy loss and Kullback-Leibler divergence loss.

3.3 Model Architectures

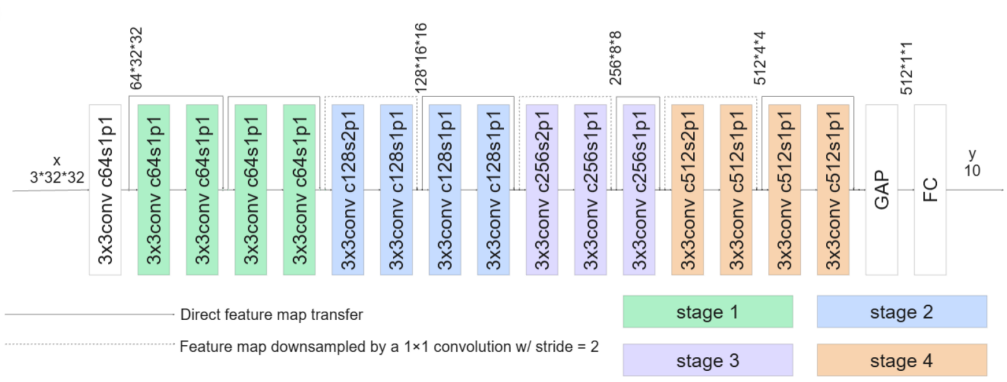


Figure 3.4: Architecture of the baseline model - modified ResNet-18 (He et al.) [5]

For my server model, a state-of-the-art model commonly found in research is used - ResNet-18 (He et al.) [5]. It's a Convolutional Neural Network (CNN) made of 17 convolutional layers and 1 fully connected layers - total 18 weight layers. It uses residual

blocks with skip connections so that gradients can go directly through the network. This helps strongly with the vanishing gradient problem.

The architecture of the ResNet-18 is as follows (figure 3.4):

- **Initial Convolution (Stem):** 7x7 convolutional layer (stride=2) at the beginning followed by a 3x3 max pooling layer. Due to the small sizes of the images in the dataset used for the experiments, the first convolutional layer was modified to avoid downsampling images too quickly: the first layer was replaced with a 3x3 convolution with stride=1 and padding=1, and the initial max pooling layer was replaced with an identity function. This modification was inspired by GitHub user Deepmancer's ResNet-18 implementation [1].
- **Residual Stages (1-4):** Each of the four stages contains two residual blocks, and each block contains two 3x3 convolution layers.
  - *Stage 1:* 64 channels with identity shortcuts (solid arrows), meaning the input is added directly to the output of the convolution block.
  - *Stages 2-4:* each consecutive stage doubles number of filter channels (to 128, 256, and 512, respectively) while halving the dimensions using stride=22. To handle this mismatch in dimensions during the skip connection, a 1x1 convolutional projection with stride=2 is used, represented by the dashed arrows in the diagram.
- **Classification Head:** a Global Average Pooling (GAP) layer is used to flatten the spatial dimensions into a 512-dimensional vector, which is then sent into a Fully Connected layer. The output is the logits for the classes in the dataset.

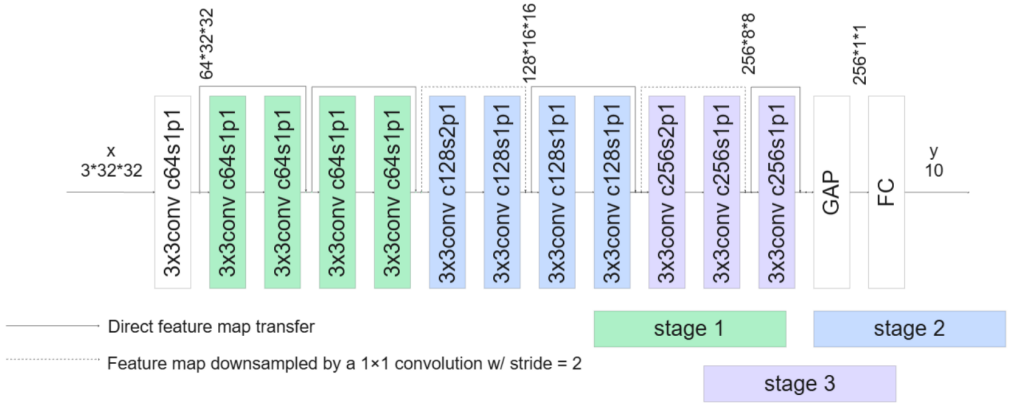


Figure 3.5: Architecture of the model ResNet-14

The architecture used for the client model is ResNet-14 (figure 3.5). It's created by removing the fourth residual stage of ResNet-18, thus reducing its size (measured in number of parameters) to about quarter of the ResNet-18's size (2.78×10^6 vs. 11.17×10^6). According to Mirzadeh et al. [20], knowledge transfer is most efficient when the student model shares structural similarities with the teacher, as well as when the difference in model sizes is not large, which is why a model based on the ResNet-18 architecture is chosen as the client model.

3.4 Dirichlet Distribution.

To answer the research questions, experiments with heterogeneous data need to be performed. Dirichlet distribution (Hsu et al.) [7] is a commonly used technique for data heterogeneity simulation. Using this distribution, a dataset can be split into unevenly distributed, non-IID subsets. For each class c, a proportion vector p_c is sampled from Dirichlet distribution:

$$p_c \sim \mathrm{Dir}(\alpha \mathbf{1}_K)$$

- α is the concentration parameter, and determines the level of data heterogeneity (lower = more heterogeneous).
- $\mathbf{1}_K$ - vector of 1s, K - number of clients and length of the vector

3.5 Dataset.

CIFAR-10 (Krizhevsky et al.) [14] is the only dataset used for all training and experiments performed. It was chosen due to its prominence in federated learning literature, where it has become a benchmark. The dataset consists of 60000 32 × 32 RGB images and 10 classes (airplane, automobile, bird, cat, deer, dog, frog, horse, ship, truck). Each class contains 6000 images and each image belongs to only 1 class.

CIFAR-10 is typically partitioned into a test set of 10000 images and a training set of 50000 images. Due to the specifics of the FedDF and BiFedDF frameworks, however, additional partitioning is needed.

FedDF and BiFedDF use an unlabeled proxy dataset during the server distillation fusion stage to get logits for training. An evaluation set is also used to measure global and local test accuracies at the end of communication rounds. The 10000 image test set is split into a 5000 image unlabeled proxy set and a 5000 image evaluation set.

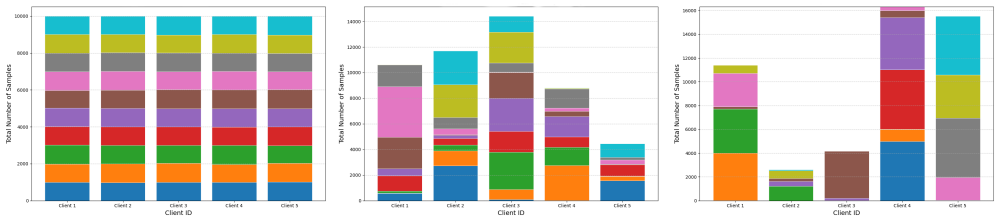


Figure 3.6: Example of the three scenarios of client data distribution: Left - IID, Middle - Non-IID Mild (Dirichlet $\alpha = 0.5$), Right - Non-IID Strong (Dirichlet $\alpha = 0.1$). *Exact class allocations vary across the 5 experimental seeds*

The client datasets use the 50000 image training set split across 5 devices in 3 different data heterogeneity scenarios (figure 3.6):

1. IID (Independent and Identically Distributed): 5 equal subsets. Each client receives 10000 samples with 1000 samples per class.

   The exact number of samples and the severity of class imbalance per client changes with the seed. The evaluation across 5 seeds is done to ensure that the observed performance is not a result of an unlucky split.

2. Non-IID: Data is split across the 5 clients using the Dirichlet distribution. Two scenarios - Mild ($\alpha = 0.5$) and Strong ($\alpha = 0.1$).

Multiple preprocessing techniques are applied to the data in order to increase generalization of the models and reduce the risk of overfitting on the dataset during the model training phase:

- Random Cropping: images are padded by 4 pixels on all sides and randomly cropped to 32×32 pixels.
- Random Horizontal Flipping: images are flipped horizontally with a 50% probability.
- **Normalization:** after conversion to PyTorch tensors, image tensors are normalized using standard CIFAR-10 dataset channel means $(0.4914, 0.4822, 0.4465)$ and standard deviations $(0.2023, 0.1994, 0.2010)$.

3.6 Experimental Setup

All experiments are performed in a simulated environment on a single desktop PC with an RTX 5080 GPU, AMD Ryzen 9 9900X CPU, and 32 GB RAM. The frameworks are implemented using Python and the PyTorch [21] deep learning library.

The server (ResNet-18) and client (ResNet-14) models are pre-trained to establish a strong baseline.

- ResNet-18 is pre-trained for 20 epochs using Cross-Entropy loss and Adam optimizer (learning rate = 0.001, weight decay = 0.000125). A Cosine Annealing learning rate scheduler is used as well. The accuracy of 92.55% is achieved with this setup.
- ResNet-14 is pre-trained with knowledge distillation for 50 epochs. The pre-trained ResNet-18 is used as a teacher, and the loss function is a combination of Kullback-Leibler Divergence Loss (temperature T = 2.0) and Cross-Entropy loss defined earlier (function weights $\beta = 0.5$). Adam optimizer is used. The accuracy of 87.64% is achieved with this setup.

For the federated learning simulation, the server and the client models were initialized with the pre-trained weights, so the round 0 server accuracy is 92.55%, and 87.64% for clients. This is done in order to isolate the variables and measure how the bidirectional component of BiFedDF and the level of data heterogeneity affect the fusion gap.

The federated learning simulation lasts for 70 communication rounds. There are 5 client models and all 5 participate in the training process each round. The batch size is 512 for both client and server data loaders.

Client models are trained for 3 epochs every round on their local data. Stochastic Gradient Descent (SGD) optimizer is used (learning rate = 0.005, momentum = 0.9). FedDF uses Cross-Entropy loss, while BiFedDF uses a combined Cross-Entropy Loss and KL-Divergence loss (temperature T = 2.0).

The server model is trained via knowledge distillation on the proxy dataset for 3 epochs per round. Both FedDF and BiFedDF use an SGD optimizer (learning rate = 0.0001, momentum = 0.9) and KL-Divergence loss (temperature T = 2.0).

The 6 FL scenarios are performed under 5 different seeds and averaged for most of the plots.

3.7 Evaluation Metrics

To ensure consistency across models, all accuracy evaluations are performed with a 5000 image evaluation dataset.

The main classification metrics are:

- **Global Server Accuracy ($Acc_s^{(t)}$):** The classification accuracy of the fused global server model (ResNet-18) at round t averaged across the 5 seeds.
- **Average Client Accuracy ($\frac{1}{K}\sum_{k=1}^{K} Acc_{c,k}^{(t)}$):** The mean classification accuracy across all $K = 5$ local client models (ResNet-14) at round t averaged across the 5 seeds.
- **Per-Class Accuracy Heatmaps**: a visualization of the round 70 classification accuracy for each of the 10 CIFAR-10 classes averaged across 5 seeds. 2 plots per each of the 6 setups: one for server accuracy, one for averaged client accuracy.

The per-class accuracy heatmaps will help us see how the federated learning approaches deal with catastrophic forgetting.

To measure the fusion gap, the following metrics are employed:

- **Server-Client Difference (Δ):**

$$\Delta_f^{(t)} = Acc_s^{(t)} - \frac{1}{K}\sum_{k=1}^{K} Acc_{c,k}^{(t)}$$

  Where $Acc_s^{(t)}$ - server accuracy at round t, $Acc_{c,k}^{(t)}$ - client k's accuracy at round - initial server accuracy. Accuracies are averaged across the 5 tested seeds.

- **Global Server Degradation (δ_s):**

$$\delta_s^{(t)} = Acc_s^{(t)} - Acc_s^{(1)}$$

  Where $Acc_s^{(t)}$ - server accuracy at round t, Acc_s^1 - initial server accuracy. Accuracies are averaged across the 5 tested seeds.

- **Average Client Degradation (δ_c):**

$$\delta_c^{(t)} = \left(\frac{1}{K}\sum_{k=1}^{K} Acc_{c,k}^{(t)}\right) - \left(\frac{1}{K}\sum_{k=1}^{K} Acc_{c,k}^{(1)}\right)$$

  Where $Acc_{c,k}^{(t)}$ - client k's accuracy at round t, $Acc_{c,k}^1$ - initial accuracy of client k. Accuracies are averaged across the 5 tested seeds.

Round 1 is used as the baseline for degradation plots to isolate the federated training dynamics from the initial catastrophic shock caused by the first exposure to skewed local datasets.

The server-client difference plot allows us to look at the relationship between the server and client's accuracy for FedDF and BiFedDF, while the two separate degradation plots will show us the stability of the models.

To examine the server behavior better, the following plots are also used:

- **Pairwise Prediction Agreement:** consists of 6 heatmaps (one for each FL setup, accuracies averaged across 5 seeds) and shows how likely the clients are to output the same class predictions on the test set (agree with each other). High agreement shows loss of diversity.

- **Ensemble Gain:** accuracy of the averaged client logits minus averaged accuracy of individual clients (accuracies averaged across 5 seeds). Lower gain shows a lack of diversity needed for a good consensus.

Chapter 4

Experiment Results

The experiments are conducted under the setup described in the previous section - 70 communication rounds, 6 scenarios (FedDF and BiFedDF frameworks with IID, Mild Non-IID and Strong Non-IID data distribution), and 5 seeds.

The runs under 5 seeds have all shown similar patterns, and therefore a decision to use an average of each scenario's performance across the 5 seeds was made for most of the plots.

4.1 Server vs Client Performance

Running the experiments yielded the following results:

Table 4.1: Baseline accuracies results

Server accuracy (ResNet-18)	**Client accuracy (ResNet-14)**
92.55	87.64

Table 4.2: Experiment results (averaged across 5 seeds)

Federated system	**Server accuracy**	**Avg client accuracy**
`FedDF IID`	90.06	85.14
`FedDF Mod Dirichlet`	89.14	78.62
`FedDF Strong Dirichlet`	86.83	54.81
`BiFedDF IID`	89.03	86.14
`BiFedDF Mod Dirichlet`	87.44	82.42
`BiFedDF Strong Dirichlet`	79.74	70.76

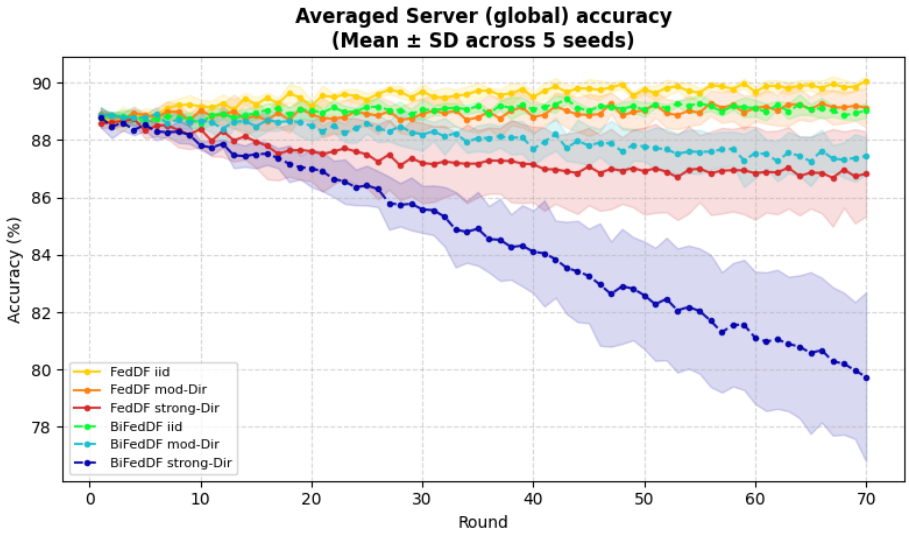


Figure 4.1: Experiment results: server model accuracies for each FL round

Overall, non-bidirectional FedDF outperforms the bidirectional approach when it comes to server accuracy (table 4.2). The final accuracies were noticeably greater than those of the bidirectional approaches. The server accuracy per round plot (figure 4.1) also shows that FedDF stabilized faster than BiFedDF in the long run. BiFedDF had much greater struggles with data heterogeneity and failed to stabilize as training continued.

One thing that is common for all methods, however, is that all of the methods result in a performance drop for the pretrained model. Some methods are more stable than others and regain some of the lost performance, but overall the performance drops from the start value.

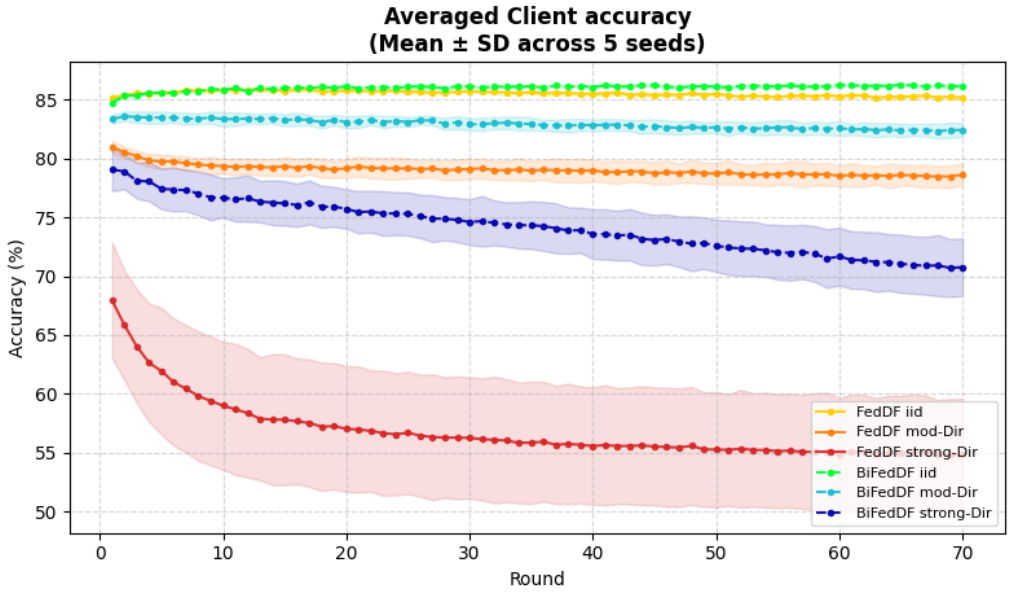


Figure 4.2: Experiment results: mean client accuracies for each round

The bidirectional approach yielded significantly better mean client accuracies (figure 4.2), outperforming FedDF in all 3 scenarios. On top of that, BiFedDF resulted in not only significantly better, but slightly more stable accuracy than FedDF on highly non-IID data, where it can be seen how FedDF clients quickly lose performance and end up having the average accuracy of 54.81%, and BiFedDF clients, despite gradually losing performance, behave more predictably than FedDF and end up having an accuracy of 70.76%.

4.2 Fusion Gap

The plots used to analyze the Fusion Gap are the server-client difference line chart and the degradation bar chart. Examining them will help us answer the research questions defined in section 1.

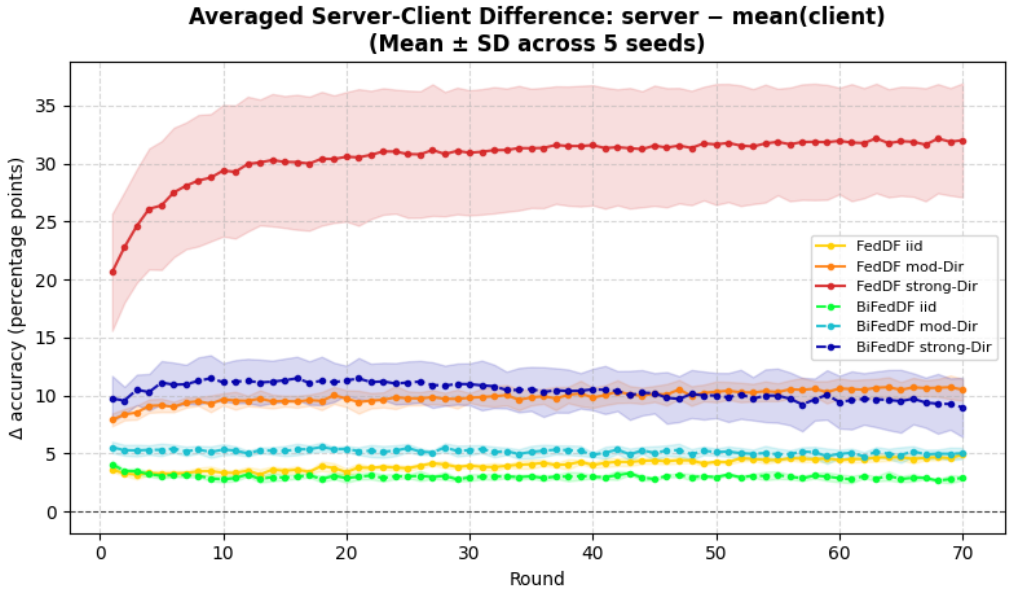


Figure 4.3: Experiment results: server-client difference for each round (averaged across 5 seeds)

According to the server-client difference plot (figure 4.3), BiFedDF ourperforms FedDF in all 3 data heterogeneity scenarios when it comes to server-client difference. Especially with high data heterogeneity, where FedDF experiences a large 30% gap between client and server performance, while BiFedDF reduces the difference to 10%.

It also reveals an interesting relationship between mean student accuracy and server accuracy. In IID and mild non-IID scenarios, the plots seem to show that the mean client accuracies and server accuracies are almost proportional to each other. This makes sense for FL with knowledge distillation, as KD's goal in the framework is to transfer knowledge, and this proportional relationship mainly highlights the regularization capabilities of knowledge distillation.

What is important to point out is that the difference can still be low and/or reduce as the rounds go on even if the performance of server and client models is dropping. Due to

this and due to all methods failing at maintaining or improving on the initial accuracy of the pre-trained models, a performance degradation evaluation should also be performed:

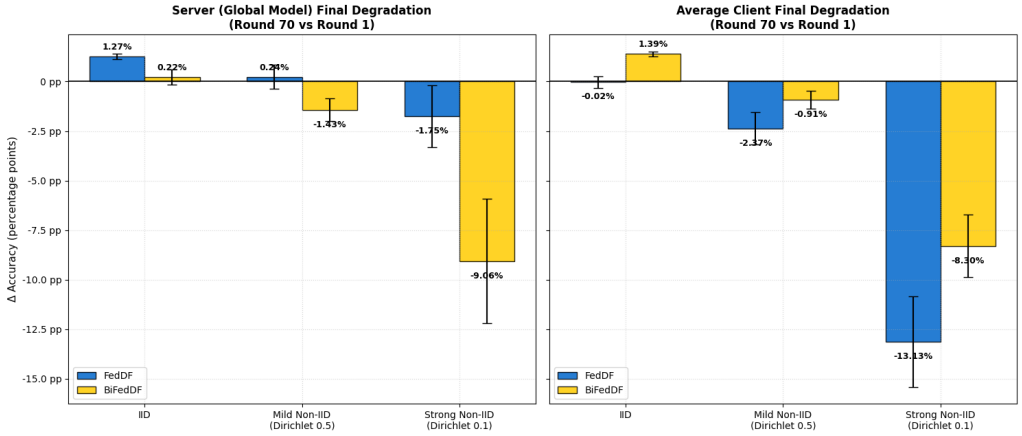


Figure 4.4: Experiment results - Degradation: Left - Server, Right - Average Client. Averaged across 5 seeds.

The degradation plot (figure 4.4) highlights a dynamic between server and client models for FedDF and BiFedDF. For FedDF, we see low server degradation at the cost of very large client degradation of up to -13.13% on Strong Non-IID data. For BiFedDF, we see a mostly opposite picture: the server performance degrades greatly (-9.06%), while client performance degrades much less (-8.30%). While it seems that the difference between client performance degradation is not that high, it must also be noted that the round 1 accuracy of FedDF (68%) is significantly worse than that of BiFedDF (79%) for high data heterogeneity (Figure 4.2). This shows that what BiFedDF really does is it essentially trades server performance for better client performance and stability.

FedDF, while failing to improve server performance with high data heterogeneity, clearly outperforms BiFedDF when it comes to achieving good server accuracy. When it comes to client accuracy, however, BiFedDF ends up being a clear winner.

4.3 Catastrophic forgetting

Local class-level dynamic can help us understand the behavior of the fusion gap under heterogeneous data distributions better. The models in this experiment are initialized with pre-trained weights, and already have high accuracy on the first FL rounds. This means that the following performance drop is not a failure to learn, but something different. Highly heterogeneous data means that models only get exposed to a subset of the total classes, and thus they overwrite the pre-trained weights that deal with classes not visible in the subsets, and thus forget how to classify the items belonging to those classes. In the work by Kirkpatrick et al. [13] this phenomenon is defined as catastrophic forgetting. The understanding of this inference dates back to 1999 (French et al.) [2]. Shoham et al. [25] connects catastrophic forgetting to my topic by showing that Non-IID data also causes catastrophic forgetting in federated learning.



Figure 4.5: Experiment results: Per-Class Server Accuracy Heatmaps for FedDF under 3 different data heterogeneity scenarios (Averaged across 5 seeds)

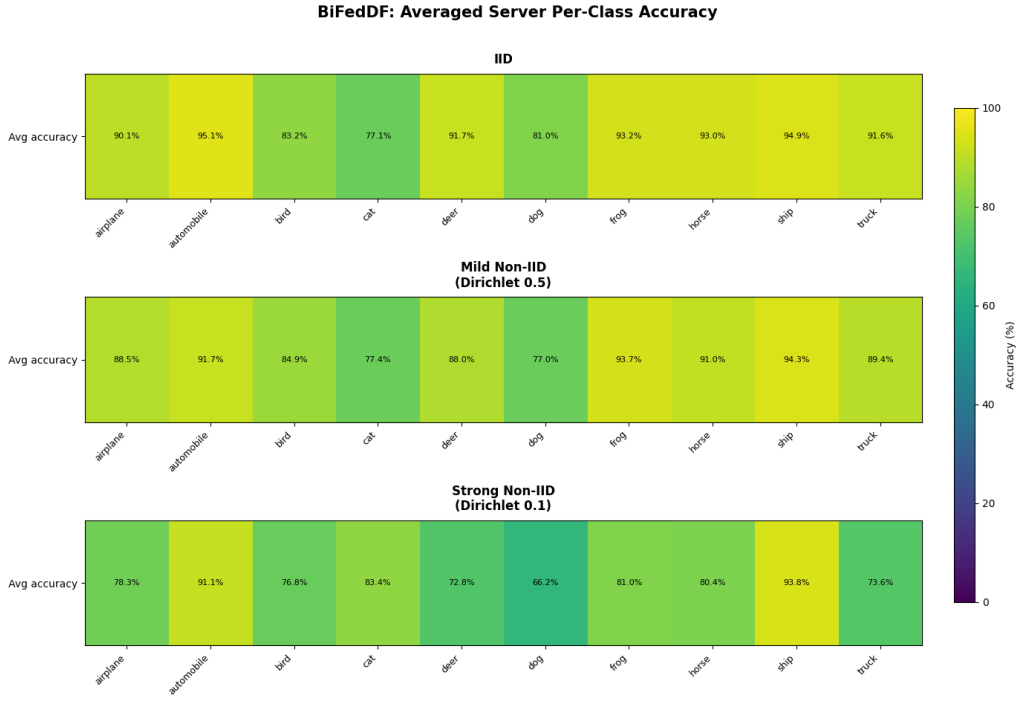


Figure 4.6: Experiment results: Per-Class Server Accuracy Heatmaps for BiFedDF under 3 different data heterogeneity scenarios (Averaged across 5 seeds)

Looking at the per-class server accuracy heatmaps for FedDF (figure 4.5) and BiFedDF (figure 4.6), we can see how the fusion gap impacts the server: the BiFedDF server degrades across most classes (e.g., "dog" drops to 66.2% and "deer" to 72.8%), while the FedDF server maintains a decent accuracy across all classes despite heterogeneous data.

Looking at the per-class server accuracy heatmaps for FedDF (figure 4.5) and BiFedDF (figure 4.6), we can see that, despite the accuracy per round plots showing FedDF to outperform BiFedDF as data heterogeneity increased, BiFedDF achieves more balanced performance across classes. More specifically, we can see how the "cat" class accuracy drops down to 70% for FedDF, but stays at 79.9% for BiFedDF under strong Non-IID data.



Figure 4.7: Experiment results: Per-Class Average Client Accuracy Heatmaps for FedDF under 3 different data heterogeneity scenarios (for a single seed)

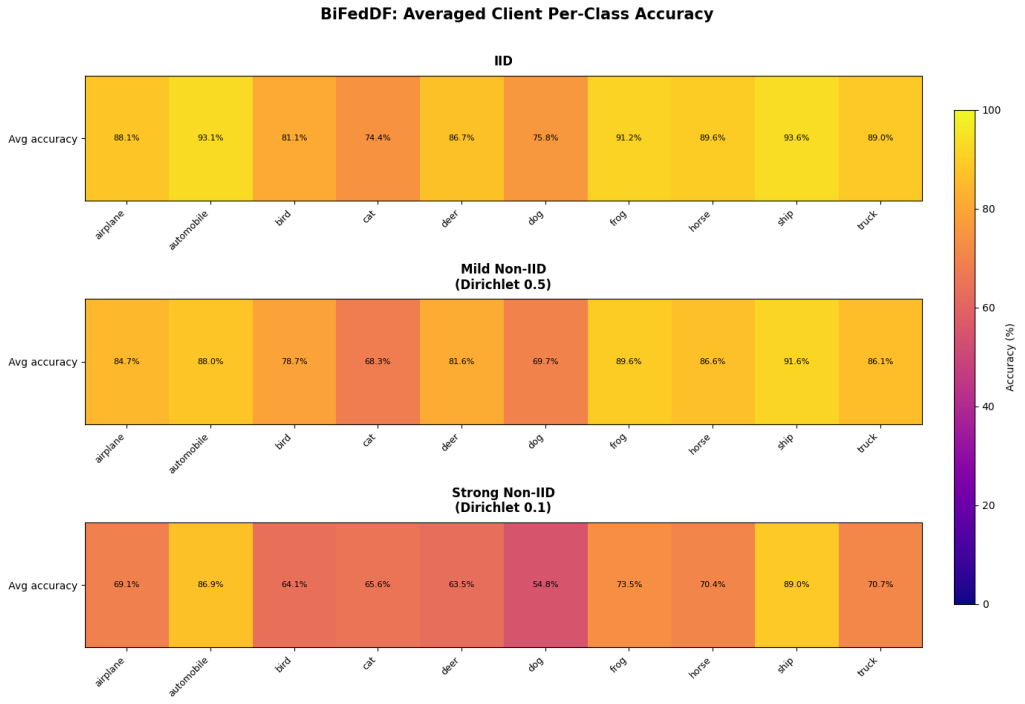


Figure 4.8: Experiment results: Per-Class Average Client Accuracy Heatmaps for BiFedDF under 3 different data heterogeneity scenarios (for a single seed)

Looking at the average per-class client accuracy heatmaps for FedDF and BiFedDF (figure 4.7 and 4.8), we can see that an overall better and more balanced performance can be observed for the client accuracy heatmaps, and here BiFedDF outperforms FedDF in all categories under strong Non-IID data. This is a good example of server's regularization at work in BiFedDF

Based on what can be seen on the per-class accuracy heatmaps, we can make the following conclusions:

- BiFedDF results in more severe server per-class accuracy drops, but more balanced per-class accuracies as data heterogeneity gets more severe. This doesn't apply for average client per-class accuracies, however. We can see that BiFedDF results in better per-class accuracy for most to all classes as heterogeneity increases. Essentially, the server provides regularization by acting as a teacher in BiFedDF, in addition to acting as a student during fusion.
- Due to the regularization properties of the bidirectional component of BiFedDF, the clients end up learning to classify the classes they were otherwise unable to learn from their own datasets. This way, the bidirectional component mitigates forgetting, and thus reduces the difference between client and server performance, lowering the fusion gap.

4.4 Loss of Ensemble Diversity

BiFedDF succeeds at mitigating the clients' catastrophic forgetting. However, as mentioned earlier, the catastrophic forgetting mitigation comes at the cost of server performance. According to Kuncheva et al. [15], an ensemble of models yields the best results

when the models in the ensemble are diverse in their results. This is supported by neural network ensemble theory (Hansen et al.) [4], which states that aggregated predictions outperform individual networks only if their errors are uncorrelated. In knowledge distillation, this means that when using an ensemble of models as a teacher, the best results will be achieved if the models in the ensemble are diverse. Analyzing the diversity of the client ensemble can help us understand why the server degrades.

In order to analyze the diversity of the client models, pairwise prediction agreement matrices are calculated for both FedDF (figure 4.9) and BiFedDF (figure 4.10) frameworks.

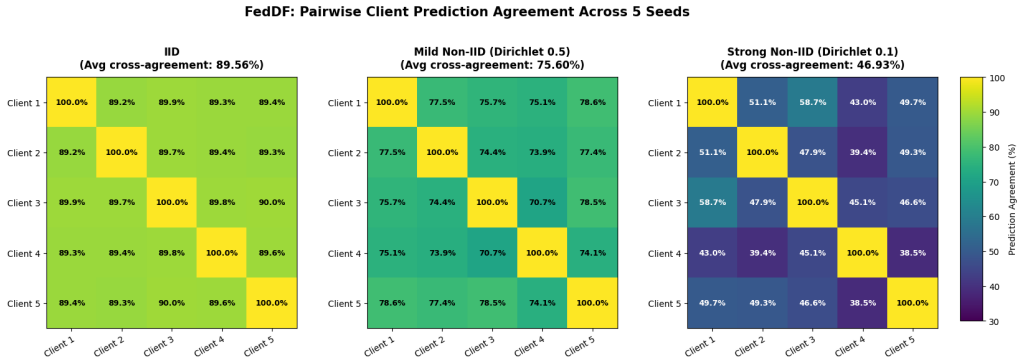


Figure 4.9: Experiment results: Pairwise Client Prediction Agreement Heatmaps for FedDF under 3 different data heterogeneity scenarios (averaged across 5 seeds)

The FedDF agreement matrices (figure 4.9) show high levels of disagreement among client models as data heterogeneity increases. The average cross-agreement percentage for strong Non-IID data is only 46.65%. What this means is that the client diversity is very high with FedDF.

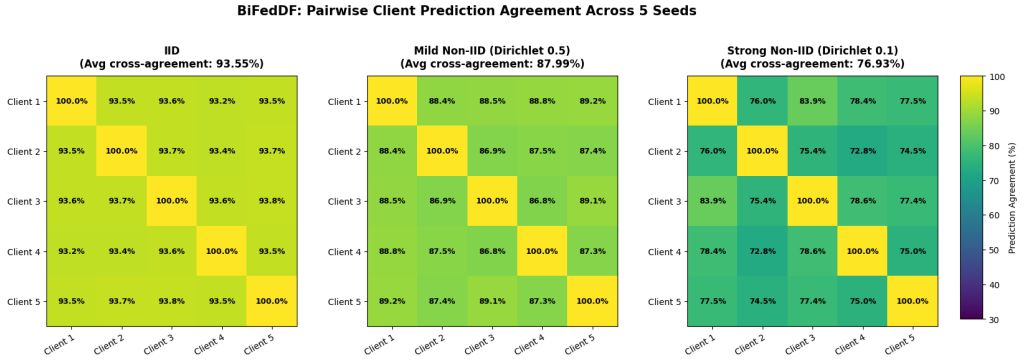


Figure 4.10: Experiment results: Pairwise Client Prediction Agreement Heatmaps for BiFedDF under 3 different data heterogeneity scenarios (averaged across 5 seeds)

On the other hand, The BiFedDF agreement matrices (figure 4.10) show an opposite picture: the agreement percentage is much higher than with FedDF for all data distributions. For strong Non-IID data, the average cross-agreement reaches 76.39% - the client models are not diverse.

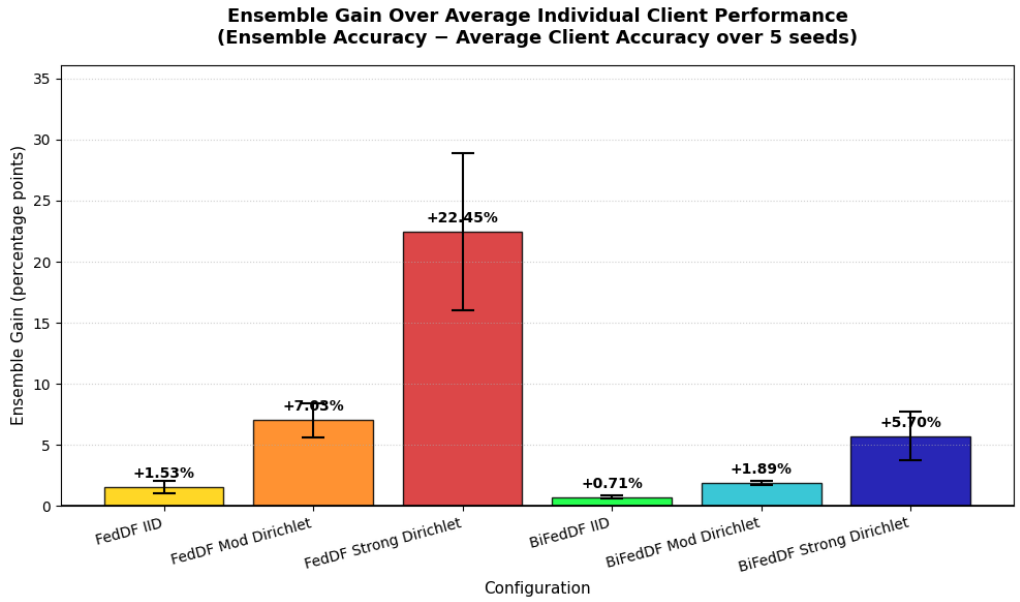


Figure 4.11: Experiment results: Ensemble Gain Over Client Performance. Averaged across 5 seeds.

The Ensemble Gain Over Client Performance plot (figure 4.11) shows how much better the performance of client models in an ensemble is than the avergaged performance of individual models. On the plot we can see that ensemble performance of clients in FedDF achieves a gain of +22.45% due to high client diversity, while BiFedDF only achieves a gain of +5.70%. During fusion, BiFedDF basically learns from its smaller copies, which ends up harming the global server model performance in the long run.

4.5 Research Questions

From the discussion and graph analysis, we can answer the research questions defined in section 1:

1. **RQ1:** Based on the experiment results and discussion, the following conclusions can be made about the effect of the degree of data heterogeneity on the fusion gap's magnitude:

   - Both the Bidirectional FedDF and regular FedDF systems struggle heavily with maintaining good server performance with highly heterogeneous data. However, the regular FedDF handles the increase in heterogeneity slightly better. Neither frameworks achieve or outperform the initial server accuracy.
   - The increasing level of data heterogeneity also strongly and negatively affects client performance and client stability for both systems, although BiFedDF is handling heterogeneous data noticeably better than the non-bidirectional counterpart. In general, average client accuracy is more heavily affected by the increase in data heterogeneity than the server accuracy.

   Looking at the fusion gap plots (figure 4.3, figure 4.4,) it can be seen that the increased model performance and decreased stability caused by data heterogeneity affects the

fusion gap: as the level of heterogeneity increases, the behavior of the fusion gap becomes less stable: it tends to grow or diminish gradually as the training goes on, and the overall fusion gap magnitude increases for both approaches. It can be concluded that the increasing level of data heterogeneity negatively affects the magnitude and stability of the fusion gap.

2. **RQ2:** The following conclusions can be made about the effect of the bidirectional distillation component's effect on the magnitude of the fusion gap:

   - Looking at the fusion gap plots (figure **??**, figure 4.4,), the fusion gap appears to be only really prevalent in BiFedDF, especially at higher data heterogeneity levels, since FedDF has an opposite issue: the client performance degrades while server remains stable. Bidirectional distillation component changes the behavior of the performance loss caused by data heterogeneity: the global server accuracy ends up being traded for local client accuracy. This is due to the bidirectional component acting as a regularizer for the clients, which increases individual client performance, but reduces diversity, which ends up harming their ensemble performance during server fusion.
   - What happens in BiFedDF is essentially a reversal of what happens in FedDF. Bidirectional distillation component ends up creating the fusion gap in the form of a server-client accuracy tradeoff.

   Overall, it can be concluded that BiFedDF actually amplifies the fusion gap, and the increasing data heterogeneity also negatively affects it by harming client and server performance for both FedDF and BiFedDF.

Chapter 5

Conclusions and Future Work

This thesis allows us to extract conclusions on the impact of data heterogeneity and bidirectional distillation in federated learning. Defining the fusion gap as a phenomenon of global server performance degradation over time while client remains stable helps us evaluate how FedDF and BiFedDF systems respond to varying levels of data heterogeneity, as well as gave us insight into the relationship between the client and the server in federated distillation.

5.1 Conclusions

The quantitative analysis of the simulated 70-round federated learning environments leads us to the following insights:

1. **FedDF Catastrophic forgetting:** The global server model maintained good accuracy during FedDF evaluation even when dealing with highly heterogeneous data. However, the client models lose performance due to unevenly distributed data, which results in catastrophic forgetting. The bidirectional distillation component in BiFedDF solves this failure by broadcasting the server's soft targets to clients, making the server model serve as a regularizer and helping the clients retain the knowledge of the classes underrepresented in the dataset.
2. **BiFedDF Diversity loss:** While BiFedDF successfully mitigates catastrophic forgetting, it does come at a cost. By mimicking the server, the client models become very homogenized under BiFedDF, and they end up acting as small replicas of the server, killing most of the mathematical client diversity, thus harming their ensemble performance. This ends up heavily damaging the server performance, especially at higher data heterogeneity levels.
3. **Nature of the fusion gap:** In conclusion, the fusion gap cannot be properly mitigated, only shifted. The FedDF framework, while technically not experiencing the fusion gap under its stated definition, struggles heavily with maintaining good client performance, but maintains server stability. BiFedDF experiences the opposite issue: the clients retain most of their performance and their stability is greatly improved compared to FedDF, while the server performance suffers greatly and keeps going down. So the real difference between FedDF and BiFedDF performance wise lies in the tradeoff between server and client performance.

5.2 Future Work

While the thesis helps us unlock new insights into knowledge distillation in federated learning, there are still areas to improve on and more work to be done:

- **Real-world hardware deployment.** All of the experiments were performed in a simulated environment. While this does help ensure absolute control over all variables, real-world hardware deployment would let us find out more about communication latency and bandwidth costs of bidirectional knowledge distillation in federated learning.
- **Analyzing the proportionality of the fusion gap.** A mathematical and theoretical analysis of the relationship between server and client performance in federated distillation systems, especially bidirectional ones, presents an interesting modeling opportunity. By finding a tie between server accuracy and ensemble diversity, models with capabilities such as, for example, server convergence limit prediction based on client agreement metrics, can be made.
- **Alternative loss functions.** The implementation of BiFedDF in this paper uses the asymmetric Kullback-Leibler Divergence loss. Symmetric metrics, such as the Jensen-Shannon divergence (JSD), used in the original bidirectional distillation paper [26], should be tested as well, as they may encourage better and more balanced weight transfer, and thus help maintain client diversity in BiFedDF.
- **Dynamic loss weighting.** The weights of the current loss function in BiFedDF are static and equal, meaning that the combined loss functions are equally as important no matter what. Dynamic loss weighting (Kendall et al.) [12], or dynamically adjusting the loss weights based on the client's local confidence or the current communication round, may prove beneficial to BiFedDF performance by finding a good balance that prevents forgetting while preserving local diversity in client models. This should be experimented with in the future iterations of BiFedDF.

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Appendix A

Ablation Study: Impact of Client Pre-training

Both the server and client models were initialized with pretrained weights for the experiments in order to isolate the fusion gap-related issues such as catastrophic forgetting, diversity and performance degradation. In this ablation study, the 6 FL scenarios are performed with only server having pretrained weights, and clients having randomly initializaed weights.

This study will help us observe the cold start behavior of the frameworks. The experiments are performed over 70 rounds.

A.1 Analysis and Insights

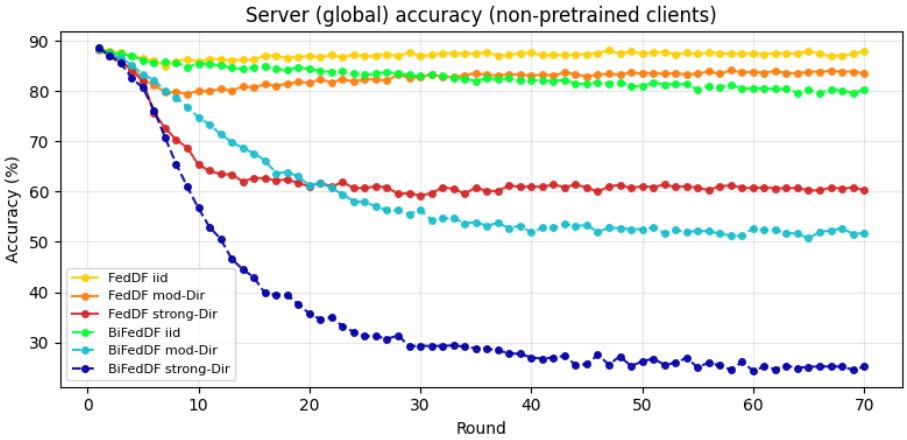


Figure A.1: Experiment results: server model accuracies for each FL round (without pretrained clients)

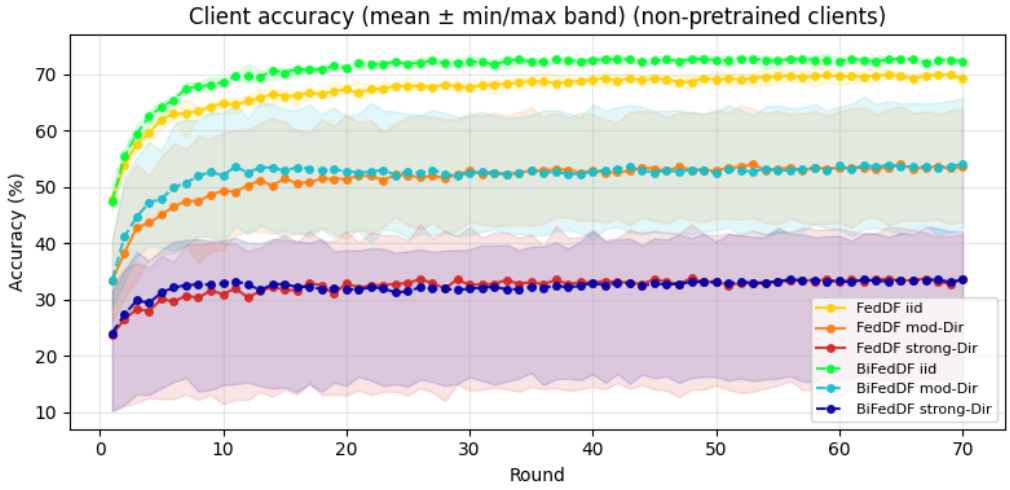


Figure A.2: Experiment results: mean client accuracies for each round (without pre-trained clients)

The server accuracy plot (figure A.1) shows how the server performance degrades way harder for all FL frameworks compared to the pretrained runs, especially in most of the initial rounds. This makes sense, as when the server distills knowledge from the consensus of randomly initialized, barely trained clients, the trained weights get overwritten by noise. As the clients are randomly initialized, the first few rounds of training are not going to bring good results immediately, so the performance suffers.

BiFedDF ends up completely collapsing under strong non-IID conditions (figure A.1), dropping from the accuracy of 88% to 25%. This can be explained with the following feedback loop that occurs in this experiment:

- Untrained clients generate a poor consensus (figure A.2).
- Server learns from a poor consensus and degrades.
- Degraded server is broadcasted back to the clients to act as their teacher.
- The degraded representations from a teacher serve as a ceiling that prevents the clients from learning effectively from local data

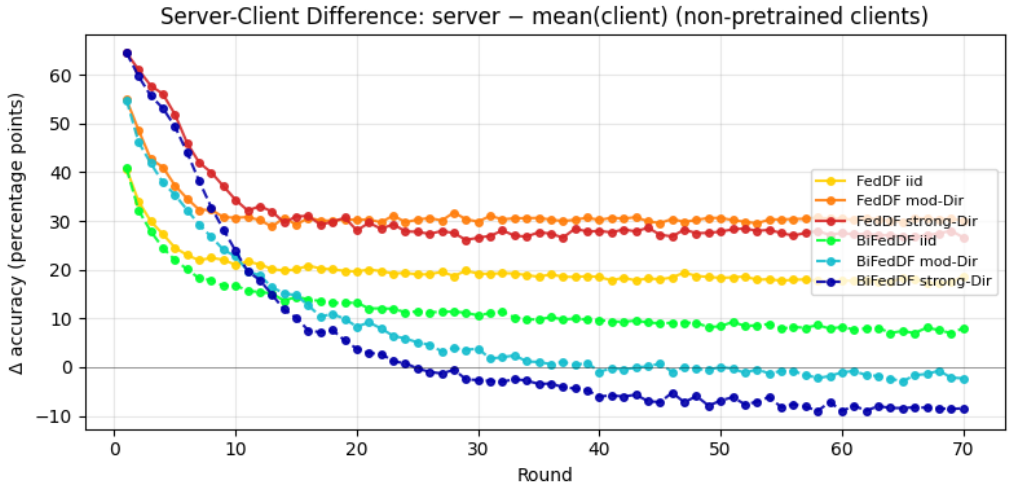


Figure A.3: Experiment results: server-client difference for each round (without pre-trained clients)

Figure A.3 shows that the mathematical difference between server and clients goes to zero or outright flips for BiFedDF as data heterogeneity increases. In combination with figures A.1 and A.2, this shows that the server collapses to the low performance level of the clients, and going as far as to perform worse than the client ensemble.

A.2 Conclusion

This ablation study confirms that distillation-based federated learning frameworks (especially bidirectional ones) are highly vulnerable to cold-start problems under heterogeneous constraints. An untrained ensemble of students prevents meaningful knowledge transfer to the server and causes the server to instead unlearn its own features. This validates the methodological decision to initialize all models with pre-trained weights, ensuring the main study measured actual heterogeneous data constraints rather than cold-start initialization failures.